

Home Interior Design Suggestion System using Deep Learning

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Abstract— This research investigates the transformative potential of deep learning in revolutionizing home interior design through the development of a web application with three core functionalities: object replacement within an existing interior design image, object removal from the interior design image, and text-to-image design generation based on user inputs. Leveraging deep learning models like ControlNet, DeepFill, and stable diffusion, as well as YOLOv8 for object detection and image classification (home interior/non-home interior), the proposed system is an innovative approach to democratizing design knowledge and expediting interior design procedures. This research advances interior design practices and shapes the future of personalized and effective home interior design solutions by offering effective design modifications and design suggestions.

Keywords— *home interior design, stable diffusion, text to image generation, image inpainting*

I. INTRODUCTION

The field of interior design for homes is at the nexus of creativity and utility, where the layout of rooms, furnishings, and décor work together to create settings that reflect personal preferences and improve quality of life. The emergence of deep learning technologies has brought about revolutionary changes in the interior design field, offering creative ways to enhance human creativity and optimize design workflows. The proposed research introduces a sophisticated design suggestion system (web application) that leverages three distinct types of interior design recommendations: replacing existing objects in the interior image with user-specified alternatives (image-to-image generation), removing unwanted objects from the interior image (image-to-image generation), and generating new interior designs based on text inputs (text-to-image generation).

A considerable proportion of homeowners (54%) intend to undertake Do-It-Yourself (DIY) home improvement projects, motivated by the desire for cost-effectiveness and customization [1]. However, this attempt is hampered by a lack of expertise in interior design. It is ineffective and time-consuming to browse general designs online, and they are frequently inappropriate for their particular living areas. Furthermore, many people find it difficult to modify their living spaces due to the high cost of professional interior

design services [2]. Designers have a heavy burden, and their productivity is further diminished by the frequent modifications. Creating aesthetically pleasing designs within tight deadlines remains a challenge for both designers and homeowners. Therefore, addressing the issues of low efficiency and a lack of aesthetic appeal in interior design is crucial [3].

The proposed interior design recommendation system offers three key functionalities. First of all, it uses deep learning to substitute objects in house interior images, making it simple for users to experiment with different decor items and furniture. Secondly, it deals with object removal, improving visual aesthetics by eliminating clutter and unwanted objects. Finally, the system uses cutting-edge technology to expedite the design process by producing photorealistic interior designs from text inputs.

Building upon existing research in AI-driven interior design, such as works on integrating aesthetics and efficiency, sustainable design, and text-to-image generation for this specific domain [3-9], this research proposes a novel system that empowers both homeowners and designers with advanced tools for creating and visualizing their dream spaces. The following sections will delve deeper into the literature review, methodology of the proposed system, and an evaluation of its effectiveness in transforming the interior design experience.

II. LITERATURE REVIEW

Artificial Intelligence (AI) is being used more and more in the interior design industry to improve productivity and stimulate creativity. Several studies explore AI-driven design generation. In research [3], diffusion models for effective and visually appealing interior design are presented. Text descriptions are used to generate designs quickly while taking functionality and styles into account. A novel approach to interior design creation using text descriptions is presented in the study [8], which makes use of diffusion models to boost productivity and creativity. In a similar vein, Chen et al. presents an AI technique that enhances design efficiency and quality by producing architectural designs in response to textual stimuli [10].

Some research focuses on automating specific design tasks: Jumming Chen et.al. investigates a sustainable interior design approach using intelligent design and automated manufacturing using Grasshopper (GH), emphasizing AI's potential to improve sustainability, efficiency, and quality [4]. The research [6] presents HyNet, as a deep learning approach for effective interior design texture retrieval, aiming to improve productivity through AI-powered recommendations. The examination of design aspects based on data is another area of research. In the study [5] uses a data-driven framework to examine color-material furnishing pairings in interior design to establish guidelines for these pairings via image analysis. A novel workflow using AI to automate interior design processes is presented in the study [9]. This workflow involves using image data to perceive the design environment and using a Markov Decision Process (MDP) to make design decisions. The MDP includes steps for modeling the interior space, building a Building Information Model (BIM), and applying texture data. The study [11] introduces a virtual dressing room application, offering users an opportunity to select and buy clothes virtually. This study underscores the importance of virtual and augmented reality in design processes, providing insights into user experiences and interactions in a virtual environment. The application of virtual experiences aligns with the proposed research's web app, which aims to offer an immersive and interactive design suggestion system.

Specialized text-to-image models for specific domains are introduced in additional research. A text-to-image model called iDesigner is introduced in study [7], with remarkable results in producing interior designs that are contextually accurate. Studies have also looked into the human element of text-to-image conversion. Study [12] uses text-to-image systems to investigate human-AI collaboration in the creation of art, examining how both artists and non-artists engage with AI-generated art. Research also explores image-to-image translation tasks. The efficiency of conditional adversarial

networks in image-to-image translation tasks is examined in the study [13].

Numerous studies concentrate on inpainting methods for object removal from images. LatentPaint, a novel method for image inpainting using Diffusion Models (DM) that achieves state-of-the-art results while maintaining fast inference, is introduced in the study [14]. Study [15] suggests an approach to image inpainting that combines texture-guided structure reconstruction with structure-constrained texture creation. A novel method for image inpainting is shown in Study [16], with an emphasis on concurrently recovering textures and structures. To address issues with attention-based inpainting techniques, a study [17] suggests an attention-free generator that operates more effectively. In the study [18], PD-GAN is presented as a Probabilistic Diversified GAN for image inpainting, producing a variety of realistic and varied outcomes for images that have missing regions.

Other works investigate object replacement methods in image editing. Study [19] introduces Imagic, a text-based real image editing approach for complex edits, enabling diverse edits with just one image and a text prompt. HyperStyle, a method for utilizing StyleGAN to edit real images and achieve great editability while preserving reconstruction accuracy, is presented in the study [20].

In conclusion, while the majority of the literature on AI-driven interior design focuses on sustainability, efficiency, and image editing, there is a noticeable gap in providing comprehensive solutions for personalized interior design suggestions for homes. By presenting a novel approach that includes essential elements of object removal from the existing interior design, object replacements from the existing interior design, and user preference-based design generation from the user inputs, the suggested research seeks to close this gap. The objective of the research is to improve interior design efficiency and user experience by utilizing recent advances in artificial intelligence.

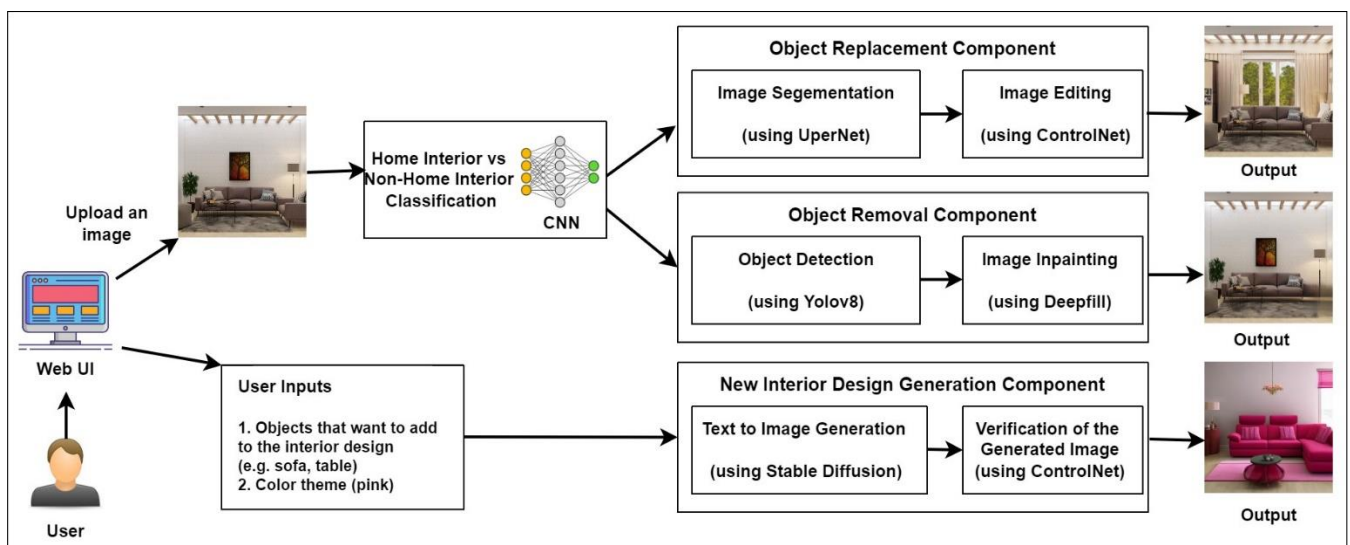


Fig. 1. System Overview Diagram

III. METHODOLOGY

The home interior design suggestion system (Fig. 1) was developed in the form of a web application. Python was used to create the web application's backend, together with FastAPI, and Uvicorn, an Asynchronous Server Gateway Interface (ASGI) server. A popular React framework called Next.js was used to develop the frontend.

When a user uploads an image, the system first classifies the image to see if it shows the interior of a house or not. Machine learning models that have been trained to distinguish between images of home interiors and other types of images are used to complete this classification task. The user can receive design recommendations if the supplied image is identified as a home interior image; if not, a suitable message prompts the user to upload a home interior image.

The web UI presents the user with three options from which to choose their preferred design type. Three types of interior design suggestions are offered by the system: replacing existing objects in the interior image with user-specified alternatives (image-to-image generation), removing unwanted objects from the interior image (image-to-image generation), and generating new interior designs based on text inputs (text-to-image generation).

A. Home Interior vs. Non-Home Interior Classification

The objective of the classification task was to distinguish between images showing home interiors and those featuring non-home interior scenes. Each class in the dataset included 1000 images. The dataset for home interior images included a variety of situations, including living rooms, bedrooms, kitchens, and so on, while the dataset for non-home interior images included a wider range of images that did not include interior settings.

The dataset was split into training (80%), validation (10%), and testing (10%) sets. Using the ImageDataGenerator library from Keras, data augmentation techniques were used for the training dataset. These methods, which included scaling, horizontal flips, and random rotations, were intended to increase the training data artificially and enhance the generalization skills of the model.

For the classification, five pre-trained Convolutional Neural Network (CNN) architectures were assessed: VGG16, ResNet50, InceptionV3, MobileNetV3Large, and DenseNet121. Utilizing pre-trained weights from the ImageNet dataset as a foundation for fine-tuning the house interior/non-home interior classification, all models made use of transfer learning. For model architecture customization, the base model layers were frozen to prevent overfitting to the limited dataset. The base model was extended with custom classification layers that included a flattening layer, a dense layer activated by ReLU, a dropout layer, and a final dense layer activated by sigmoid. After the model was constructed, it was trained for 50 epochs using the Adam optimizer with a learning rate of 0.0001.

B. Replace Existing Objects in the Interior Image with User-Specified Alternatives

The uploaded image is segmented using a pre-trained deep learning model, called UpernetForSemanticSegmentation. Using color, segmentation partitions the image into distinct regions. The result is a segmentation map, where each pixel is assigned a label corresponding to its belonging object class (e.g., sofa, chair, table). A binary mask is produced using the segmentation map as a basis. In the segmentation map, the target object's matching pixels are set to 1, while all other pixels are set to 0. This mask isolates the specific object region for manipulation.

A pre-trained ControlNet model is employed for image generation. ControlNet is a type of GAN architecture specifically designed for image editing tasks. Three pieces of information are fed into the ControlNet pipeline: original image, object mask, and segmentation map.

Text prompts are provided to guide the image generation process. The desired result (e.g., "an image of a room, interior design, 4K, high resolution") is described in the positive prompt. The creation of an image that mimics an interior design snapshot is encouraged by this prompt. Unwanted image characteristics (such as "low resolution, watermark, blurry") are listed in the negative prompt. This helps prevent artifacts and maintains image quality.

ControlNet creates a new image with an alternative for the target object based on the input information. The user receives the final image, in which the target item has been substituted, as the output.

C. Remove Unwanted Objects from the Interior Image

To identify objects in the user uploaded image, the object detection model is applied. The YOLOv8 medium variant (YOLOv8m) is used to train the object detection model using a dataset of 475 home interior images. YOLOv8m is selected due to its capacity to detect objects in real time and its appropriateness for implementation in web-based applications. This dataset contains a wide variety of interior settings with different pieces of furniture and décor. The model was trained to identify 13 classes of frequently encountered items in interior design, such as tables, chairs, sofas, and plants.

Using the user interface, the user selects the undesirable object to be removed by providing its class label (for example, "sofa"). The undesired object in the image is isolated by a binary mask that the system creates using this information. By locating the bounding box connected to the user-specified class label, this mask is produced. All other regions of the mask stay black, while the corresponding area of the original image is set to white.

Based on the created mask, an inpainting model (deepfill) eliminates the undesired object. The input for the inpainting

model is the mask as well as the original image. It makes use of the data from the surrounding image areas to forecast and complete the space where the undesired object was supposed to be.

D. Generate New Interior Designs based on Text Inputs

To convert user input (text prompts) into interior design images, the system makes use of a pre-trained text-to-image diffusion model. Both a positive and a negative prompt are included in the prompt's construction. The desired scene is fully described in the positive prompt. It includes phrases like "a photograph of a room" and specifies the objects the user wants to see ("including sofa, chair, coffee table") and the preferred color theme (pink color). It also uses the term "interior design" to indicate the stylistic intention. Terms like "4K" and "high resolution" encourage high resolution and quality. The model is guided away from undesirable outcomes with the use of the negative prompt. It uses words like "watermark," "low resolution," and "blurry" to make sure the image is clear and presents itself in a professional manner.

A pre-trained Stable Diffusion model is employed for text-to-image generation. Stable Diffusion is a powerful diffusion model capable of producing high-quality and creative images from text descriptions. The selected model variation uses mixed precision training (torch.float16) for effective inference on a GPU and is loaded from the Transformers library.

The diffusion model's initial image might not exactly match the user's preferred color scheme or object positioning. The system uses a segmentation process to address this. This pipeline consists of two components: AutoImageProcessor and UperNetForSemanticSegmentation. The image is resized and converted to the proper tensor format by the AutoImageProcessor in order to prepare it for the segmentation model. By analyzing the image and applying semantic labels to various regions, the pre-trained segmentation model (UperNetForSemanticSegmentation) essentially creates a "mask" that recognizes objects and their locations inside the image.

A ControlNet model is utilized to improve the original image by utilizing the segmentation mask and the original text prompt. Original image, segmentation mask, ControlNet conditioning image and positive, negative prompts are the inputs of the ControlNet. Goal of the ControlNet is to make sure that the items in the final image match the elements that the user chose in the text prompt, and that their location and color choices complement the segmentation mask and overall color scheme. A new image that displays the user's text description of the intended interior design is the final output.

III. RESULTS AND DISCUSSION

This section of the paper discusses the results obtained for home interior vs. non-home interior image classification, object detection, and design suggestions.

Table I presents the results of the house interior vs. non-home interior classification, showing how different CNN architectures performed and the test accuracy scores that went along with them.

TABLE I. RESULTS COMPARISON OF THE HOME INTERIOR VS. NON-HOME INTERIOR CLASSIFICATION FOR THE DIFFERENT ARCHITECTURES

Architecture	Test Loss	Test Accuracy(%)
VGG16	0.0343	98.48
ResNet50	0.5946	72.22
InceptionV3	0.0410	99.49
MobileNetV3-Large	0.4728	78.28
DenseNet121	0.0012	100.0

Among the evaluated models, DenseNet121 achieved the highest accuracy of 100%, indicating perfect classification on the held-out testing set. InceptionV3 followed closely with a test accuracy of 99.49%. VGG16 also demonstrated good performance with an accuracy of 98.48%. ResNet50 and MobileNetV3Large achieved lower accuracies of 72.22% and 78.28%, respectively.

These results highlight the effectiveness of transfer learning techniques and the choice of model architecture in accurately classifying home interior and non-home interior images. With its exceptional performance, the chosen model, DenseNet121, was integrated into the system to classify user-uploaded images.

The object detection model (Yolov8) achieved a mean Average Precision (mAP50) of 0.995 and a mAP50-95 of 0.967, indicating excellent performance in detecting and localizing objects across all classes within the dataset. Furthermore, individual class performance demonstrates high precision (P), recall (R), and mean Average Precision (mAP) values, with most classes exceeding 0.99 for both P and R. A detailed breakdown of each class's performance (Model Summary) is provided in Table II.

TABLE II. MODEL SUMMARY OF THE YOLOv8 MODEL

Class	No. of Instances	Precision (P)	Recall (R)	mAP50	mAP50-95
All	3061	0.996	0.994	0.995	0.967
Sofa	526	0.998	0.997	0.995	0.991
Chair	321	0.995	1	0.995	0.984
Small Plant	330	0.998	0.97	0.995	0.909
Large Plant	342	1	0.998	0.995	0.967
TV	62	0.996	1	0.995	0.988

TV Stand	40	0.993	1	0.995	0.979
Flower Vase	157	0.994	0.992	0.995	0.944
Table	453	0.998	0.998	0.995	0.983
Floor Lamp	174	0.999	0.994	0.995	0.962
Table Lamp	83	0.997	1	0.995	0.951
Side Table	106	0.998	1	0.995	0.971
Wall Art	403	1	0.999	0.995	0.975
Dining Table Set	64	0.984	0.974	0.994	0.975

The results obtained from the three design suggestion types are presented below. The results for the first design suggestion type, which involves replacing user-specified alternatives for existent objects in interior images, are shown in Figures 2, 3, and 4. After the wall art is replaced, the input image and corresponding output image are displayed side by side in Fig. 2. Similarly, Fig. 3 displays the input image and the final output image following the wall replacement, and Fig. 4 displays the input image and the final output image after the sofa replacement.

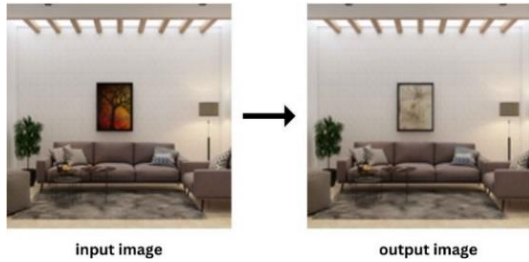


Fig. 2. Input image and the output image after replacing the wall art

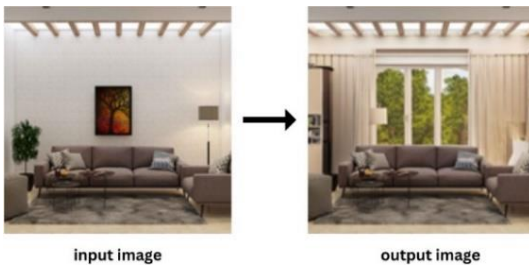


Fig. 3. Input image and the output image after replacing the wall



Fig. 4. Input image and the output image after replacing the sofa

The outputs of the second design type which is removing objects from the interior images are depicted in Figures 5, 6, and 7. In Figure 5, the input image is presented alongside the resulting output image after removing the table. Likewise, Figure 6 displays the input image and the corresponding output image after removing the lamp, while Figure 7 showcases the input image and the resulting output image after removing the chair.

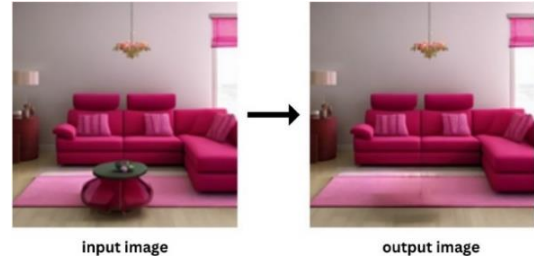


Fig. 5. Input image and the output image after removing the table

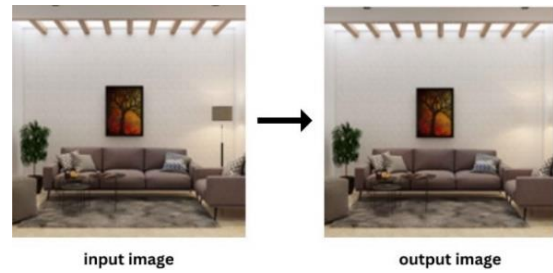


Fig. 6. Input image and the output image after removing the lamp

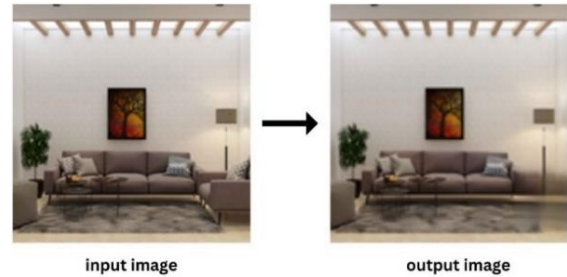


Fig. 7. Input image and the output image after removing the chair

The outputs of the third design type which is generating new interior designs based on user inputs are depicted in Figures 8, 9, 10, and 11. The outcome for the input objects—a table, sofa, lamp, and a pink color scheme—is shown in Fig. 8. Fig. 9 displays the output for the input objects, which include a wall art piece and a sofa with accent pillows and a blue color scheme. The output produced for the input objects, which included a wall art piece and a sofa with a red color scheme, is shown in Fig. 10. Finally, without defining a color scheme, Fig. 11 shows the output for input objects that include a couch, chair, table, lamp, and wall art.

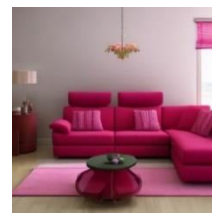


Fig. 8. Interior design for color scheme: pink

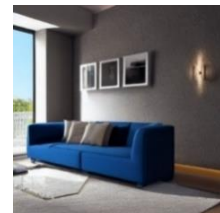


Fig. 9. Interior design for color scheme: blue

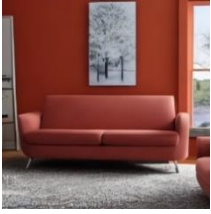


Fig. 10. Interior design for color scheme: red



Fig. 11. Interior design for without defining a color

The results that have been presented provide valuable insights into the effectiveness of the proposed deep learning-based interior design suggestion system. The system's ability to create completely new design recommendations based on user inputs, in addition to replacing and removing things from interior images, was demonstrated through the analysis of three different types of design suggestions. The illustrations presented in Figures 2 through 11 offer concrete proof of the system's capacity to improve the creative process of interior design by offering users customized and inventive design suggestions. These results demonstrate how deep learning technologies have the power to revolutionize traditional design practices and pave the way for more efficient, user-centric approaches in the field of interior design.

The proposed algorithm was validated through extensive experimentation and comparison with deep learning models. For instance, the classification model for distinguishing home interior images from non-home interior images achieved an impressive 100% accuracy with DenseNet121. Other models like InceptionV3 and VGG16 also performed well, indicating the robustness of the classification component. The YOLOv8 object detection model further demonstrated excellent performance, with a mean Average Precision (mAP50) of 0.995 and mAP50-95 of 0.967. These high-performance metrics validate the effectiveness of the algorithm in various tasks such as classification, object detection, and inpainting.

Commonly used metrics to evaluate the performance of deep learning models in interior design suggestions include accuracy, precision, recall, and mean Average Precision (mAP). For classification tasks, accuracy is a critical metric, as demonstrated in the comparison of different CNN architectures. In object detection, precision, recall, and mAP are essential metrics. The system's object detection model achieved high precision and recall across all classes, indicating reliable identification and localization of objects within interior images. Additionally, the quality of generated images for object replacement and text-to-image design generation was qualitatively assessed through visual inspection, ensuring that the results met high aesthetic and contextual standards.

User feedback plays a crucial role in refining and improving the suggestion system. By incorporating user preferences and receiving iterative feedback on design suggestions, the system can learn to tailor recommendations more effectively. This feedback loop is essential for personalizing the user experience and enhancing the system's relevance and utility. Future work will focus on integrating a feedback mechanism within the web application to gather

user responses and use this data to further train and fine-tune the models.

V. CONCLUSIONS

This research investigated the potential of deep learning for personalized and efficient home interior design suggestions. A web application was developed featuring three functionalities: object replacement within an existing interior design image, object removal from the interior design image, and text-to-image design generation based on user inputs.

The system effectively classified home interior images (home interior vs non home interior) with a DenseNet121 model achieving 100% accuracy. Other evaluated models, such as InceptionV3, VGG16, ResNet50, and MobileNetV3Large, also performed well, with test accuracies of 99.49%, 98.48%, 72.22%, and 78.28%, respectively. The YOLOv8 object detection model excelled in detecting and localizing a variety of furniture and décor items commonly found in interior scenes, achieving a mean Average Precision (mAP50) of 0.995 and mAP50-95 of 0.967. Furthermore, the system used ControlNet to replace objects in interior images, deepfill model to remove unwanted objects, and a stable diffusion model with ControlNet refinement to generate new designs based on user text inputs.

These findings highlight the effectiveness of the proposed system and the potential of deep learning for personalized and efficient home interior design. Future work could explore further refining the models and user interface even more to improve accuracy. Furthermore, increasing the object classes available for object detection would allow the system to recognize a wider variety of objects in the interior scenes. Additionally, users' preferred interior styles could be used as input to create more personalized and customized interior designs. By addressing these limitations and pursuing these future directions, the proposed system has the potential to become a valuable tool for empowering users to create dream spaces that reflect their unique styles and preferences.

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